

CLICKCRAFT AI

Production-Ready MERN Stack & Gemini AI Application

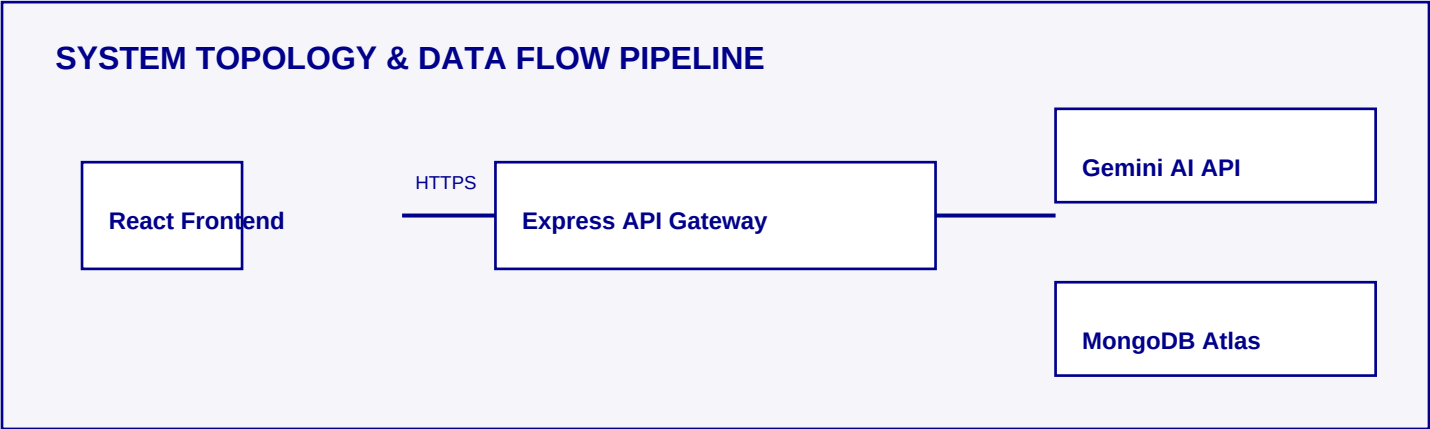
Technical System Architecture & Design Specification

Candidate / Developer:	Aman Sharma
Target Role:	Software Development Engineer (SDE)
Target Organization:	AWIROS
Application Type:	MERN Stack SaaS Platform
Core Technologies:	React 19, Node.js, Express, MongoDB Atlas, Google Gemini 1.5, Vercel
Key Architecture Patterns:	Monorepo, REST API, Serverless Edge Routing, Role-Based Access Control

This report outlines the full systems engineering, design choices, data schemas, and security implementations of the ClickCraft platform for engineering evaluation.

1. SYSTEM ARCHITECTURE & DATA FLOW

ClickCraft is a high-performance SaaS application designed to generate high-converting video thumbnails using AI. The platform features user authentication, a granular credits-based subscription system, interactive payment workflows, and an advanced folder-based file management system.



Client Layer (Frontend)

- ? Single Page Application (SPA) built with React 19 and Vite for lightning-fast bundling.
- ? State Management: React Context API for global session management, user profile data, and credits tracking.
- ? Styling & Animations: TailwindCSS for layouts and motion/react (Framer Motion) for card transition effects, modal animations, and interactive credit card validation states during pricing selection.

API & Service Layer (Backend)

- ? Node.js & Express REST API using ES Modules. Fully containerized/deployable serverless via Vercel Functions.
- ? CORS Control: Configured dynamically to match specific dev (IPv6, 127.0.0.1, localhost) and production domains.
- ? Gemini Integration: Integrates @google/generative-ai SDK. Structured prompt engineering forces Gemini to write highly optimized prompts that generate perfect context-aware thumbnail titles, descriptions, and styles.

Storage & Database Layer

- ? Database: MongoDB Atlas cloud instance with Mongoose schemas utilizing indexes on email/userId for optimized query performance.
- ? Asset Storage: Cloudinary API. The server streams thumbnail graphics directly to Cloudinary, ensuring that expensive assets are cached, delivered via global CDN networks, and don't load down our server filesystem.

2. DATA MODEL & SCHEMA DESIGN (MONGOOSE)

The relational mappings in ClickCraft are implemented using Mongoose models. User schema stores user auth state, current subscription tier, and remaining AI generation credits. Folders model category groupings, and Thumbnails refer to both owners and folders via ObjectId references.

User Model Schema	Thumbnail Model Schema
· name: { type: String, required: true } · email: { type: String, unique: true } · password: { type: String, min: 6 } · credits: { type: Number, default: 5 } · plan: { type: String, default: 'Free' } · createdAt: { type: Date } · updatedAt: { type: Date }	· userId: { ref: 'User', required: true } · folderId: { ref: 'Folder', default: null } · title: { type: String, required: true } · style: { type: String, required: true } · image_url: { type: String } · prompt_used: { type: String } · user_prompt: { type: String }

3. REST API SPECIFICATION & INTEGRATION ROUTES

POST /api/auth/register	Registers a new user, hashes password via bcrypt, allocates 5 free credits.
POST /api/auth/login	Authenticates user credentials, returns signed JWT token (expires in 30d).
POST /api/thumbnails/generate	Generates thumbnail. Enforces check: if credits < 1, blocks and returns 400 Upgrade error.
PUT /api/thumbnails/:id/move	Moves a thumbnail to a folder by updating folderId ObjectId mapping in database.
GET /api/folders	Fetches all user folders. Returns aggregates of item counts inside folders.
POST /api/folders	Creates a new named folder categories instance for the authenticated user session.
POST /api/billing/checkout	Processes checkout. Updates user credits (e.g. Pro tier) via MongoDB \$inc query.

4. SDE INTERVIEW FOCUS AREAS (AWIROS SPECIFIC)

For an SDE role at AWIROS (specialists in AI Computer Vision & Deep Tech products), focus on conveying system reliability, resource optimization, clean abstraction layers, and dynamic security controls.

A. Advanced Prompt Engineering & Pipeline Design

The AI generation module doesn't just pass raw user inputs to Gemini. It passes them through a structured pipeline that formats the user input into clear stylistic constraints. This results in stable AI outputs. In the interview, discuss how you handle API failures, write deterministic prompts, and abstract the LLM interface behind clean service endpoints.

B. SaaS Resource & Credit Limits Guards

Credits validation is implemented on both frontend and backend for defensive programming. The server verifies user credit state via Mongoose middleware before making expensive LLM calls. The billing route updates credits using atomic MongoDB queries to prevent race conditions during checkout simulation.

C. CORS Dynamic Origin Handlers

Rather than hardcoding static array limits in Express, a dynamic callback regex checks incoming origins. This allows secure, seamless requests from various development ports (IPv6 `:::1`] and IPv4 `127.0.0.1`) as well as production Vercel subdomains while keeping wildcard credentials blocked for OAuth security.

D. Performance and Clean Code Design Patterns

- ? Monorepo structure allows decoupled development of React and Express.
- ? Cloudinary CDN hosting prevents loading local server storage with huge static assets.
- ? Folders are deleted gracefully: when a folder is deleted, database hooks set associated thumbnail references back to null rather than cascading deletions, protecting user assets from accidental loss.

? PRO-TIP FOR AWIROS: Discuss how the application structures files and separates concerns. AWIROS values developers who write clean REST interfaces, optimize database queries, manage asynchronous tasks gracefully, and handle production deployment workflows (like Vercel and CI/CD) confidently.